Leakage-Aware, Sensitivity-First Validation of Blood-Smear Leukemia Classifiers: A Two-Dataset Study of Engineered Features, Deep Features and Dimensionality Reduction

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***Abstract*—Deep-learning models for leukemia detection in blood smears often report 97–100% accuracy, yet few are used in practice. We examined how much of this performance survives honest evaluation on two public datasets: 106 patients with acute myeloid leukemia (34 with acute promyelocytic leukemia, APL; 25,724 cells) split by diagnosis date, and C-NMC 2019 (73 subjects; acute lymphoblastic leukemia vs. normal B cells; 10,661 cells) with a held-out official fold. With identical data and classifiers, splitting cells instead of patients inflated the area under the ROC curve (AUROC) by 0.25–0.32 and 0.09–0.14 on the two datasets when cells carried the patient's diagnosis, but by at most 0.04 when the label described the cell. Engineered morphology features matched or outperformed frozen ImageNet embeddings in most external comparisons (blast detection 0.957 vs. 0.920; C-NMC cells 0.840 vs. 0.775), and a fine-tuned CNN was best for single-cell APL classification (0.773) but not for patient diagnosis (0.778 vs. 0.812 for attention-based multiple instance learning). Principal component analysis raised external AUROC on the 24-patient APL test (pre-declared 64 components: 0.743 to 0.819), but the gain was not significant and did not replicate in pooled cross-validation over 106 patients or on C-NMC, where development performance favored the worst choices. Thresholds fixed for 95% sensitivity transferred to C-NMC (15–17 of 17 subjects detected) but not to the later APL cohort, where they missed up to half of APL patients; recalibrating with m local APL cases yields an expected sensitivity of only $m/(m+1)$. We recommend patient-level splits, more than one external test, sensitivity-first reporting with missed-case counts, and local recalibration with about 20 confirmed positive cases before clinical use.**

***Index Terms*—Acute promyelocytic leukemia, acute lymphoblastic leukemia, blood smear, data leakage, deep learning, dimensionality reduction, external validation, feature engineering, multiple instance learning, sensitivity.**

I. Introduction

ACUTE leukemia is usually first suspected on a peripheral blood smear, where a trained morphologist looks for blasts and other abnormal white cells [1]. Acute promyelocytic leukemia (APL) shows how much rides on that first look: it is a medical emergency, because patients can bleed to death within days unless all-trans retinoic acid is started as soon as APL is suspected, often before genetic confirmation arrives. Reading smears takes time and experience, and the experience is not available everywhere, which is why automated smear analysis has attracted so much work.

Published results look encouraging. Many deep-learning models report accuracies between 97% and 100% for leukemia detection from single-cell images [2], [3], and weakly supervised models reach AUCs above 0.95 for recognizing APL [4]. Yet few of these systems are used in practice. Our starting point is that part of the gap lies not in the models but in how they are evaluated. When cells from the same patient appear in both the training and test sets, a model can succeed by recognizing the patient—the stain, the smear, the day it was scanned—rather than the disease. When the label is the patient's diagnosis copied onto every cell, this shortcut is always available. And when performance is summarized by accuracy alone, it hides the error that matters most clinically: a sick patient called healthy. Similar leakage and shortcut problems have been documented across machine-learning-based science [5] and in medical imaging [6], [7].

In this work we measure these effects directly on a public cohort of 106 patients with acute myeloid leukemia (AML), 34 of whom have APL, replicate the analysis on the C-NMC 2019 dataset of acute lymphoblastic leukemia, and build an evaluation that a clinician could trust. Our contributions are:

- C1 *A controlled leakage audit on two datasets.* With identical data, features and classifiers, we compare image-level cross-validation (CV), patient-grouped CV and an untouched external set, both for a label that belongs to the patient and for a label that belongs to the cell.
- C2 *External validation of two kinds.* Models are developed on an earlier APL-AML cohort and tested once on patients diagnosed later, and on C-NMC developed on two official folds and tested on the third; a pooled 106-patient analysis adds statistical power.
- C3 *Sensitivity-first reporting.* Decision thresholds are set on the development cohort for a sensitivity of at least 0.95, and we report recall, the number of missed APL patients, specificity, F1 score, negative predictive value and calibration, not only accuracy and AUROC. We also quantify how many local cases a new laboratory needs to restore the operating point, and the net benefit at a realistic prevalence.
- C4 *Interpretable, frozen and fine-tuned features.* We design 93 morphology features that mirror what hematologists look at, compare them with frozen and fine-tuned CNN

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features and their fusion, show through ablation which feature groups carry the signal, and test whether dimensionality reduction by principal component analysis (PCA) improves transfer, including a replication that it did not pass.

C5 *A lightweight patient-level pipeline.* Patient diagnosis is made by attention-based multiple instance learning (MIL) [8] over frozen features, trained on a CPU, and compared with established baselines on the same cohort.

Code, split files and label maps are released so that results on both datasets can be compared directly in future work.

II. Related Work

Table I summarizes representative studies. For each, the last column notes an open question that the study was not designed to answer and links it to the contribution of this paper that addresses it.

A. Cell-Level Classification of Blood Smears

Much of the literature classifies single segmented cells as leukemic or normal. The C-NMC 2019 challenge [9] set a useful standard by holding out whole subjects for testing: ResNeXt models reached a weighted F1 score of 0.889 [10], and a weighted ensemble of five CNNs reached 0.886 with an AUC of 0.941 [11]. More recent transformer and hybrid designs report accuracies of 0.97 to 1.00 on C-NMC and on the ALL image dataset [2], [3], and hybrid pipelines that feed CNN features to classical classifiers reach 0.88 when separating lymphoblasts, myeloblasts and normal cells [12]. These studies advanced architectures and training strategies. Because the split unit is not always reported, and because some of them combine sources in which class and source coincide, it remains open how much of their performance would carry over to new patients (C1, C2). A recent subject-disjoint benchmark on C-NMC found an AUROC of 0.913 [13], which suggests that the split matters, but it covered a single, cell-level label.

B. Patient-Level and Weakly Supervised Diagnosis

Clinical decisions are made per patient, and several groups have moved in that direction with MIL. Expert-level single-cell classification [14], [15] made it possible to identify genetic AML subtypes from blood-smear cells [16] and NPM1 or FLT3-ITD status from marrow [17], [18]. For APL, Sidhom *et al.* [19] released the cohort used here and showed that a single-cell model trained on the earlier patients reached a cell-level AUC of 0.890 in development and 0.743 in the later validation cohort. Manescu *et al.* [4] trained an annotation-free MIL model on the same patients and reported an AUC of 0.96 under random three-fold CV. Interpretable morphometry has also proved useful: engineered cell features with sparse regression detected myelodysplasia and anemia subtypes and held up in an external cohort [20]. Across these studies, the number of missed patients at a clinically chosen operating point and the calibration of predicted probabilities are rarely reported (C3), and engineered features have not yet been compared with deep features for acute leukemia under the same protocol (C4).

C. Foundation Models and Generalization

Hematology foundation models aim to make cell representations robust to scanners and stains. DinoBloom, trained on more than 380,000 cells from 13 datasets, outperforms generic encoders [21]; later work adapts DINOv3 with a risk-aware loss [22] and compares foundation models with retrieval under a held-out-dataset protocol [23], while LeukemiaAttri provides images of the same films from different microscopes [24]. The largest patient-level study to date trained a DinoBloom-based transformer on 6,610 individuals and validated it on two external sets, including the cohort used here [25]. Such models need large labeled cohorts and GPU resources; whether lightweight models trained on a few hundred patients can be evaluated just as rigorously is the question behind C5.

III. Materials and Methods

A. Dataset

We used the public APL-AML cohort of Sidhom *et al.* [19]. It contains single-cell images from Wright-stained peripheral blood smears of 106 adults with AML treated at Johns Hopkins Hospital between 2010 and 2020, captured with a CellaVision DM100 analyzer. Thirty-four patients have APL and 72 have other AML subtypes. The authors split the cohort by date: 82 patients diagnosed before January 2019 form the Discovery cohort and 24 diagnosed later form the Validation cohort. The 25,724 cell images measure about 360×363 pixels; 15,632 of them carry a cell type assigned by the analyzer and confirmed by its operator, and the remaining 10,092 are unlabeled. Table II and Fig. 1 describe the composition.

The cohort has several characteristics that shape the analysis. The classes are imbalanced and the imbalance shifts between cohorts: APL makes up 27% of Discovery patients but 50% of Validation patients, so accuracy alone would mean different things in the two cohorts. The number of cells per patient is confounded with the diagnosis in opposite directions—Discovery APL patients have about twice as many cells as AML patients, whereas the reverse holds in Validation—so our patient-level models are designed to be insensitive to bag size. The original authors note that all Discovery smears scanned before 2018 were APL [19], which means that acquisition date carries diagnostic information inside the Discovery cohort and invites shortcut learning whenever patients leak across splits. The mix of cell types also differs by diagnosis; labeled monocytes, for example, are nearly absent in APL. Finally, ten image files are truncated, 54 are full microscope fields rather than single-cell crops, cell labels are analyzer classifications verified by an operator rather than consensus annotations, and the Validation cohort comes from the same hospital and contains only 24 patients.

To test whether our findings hold beyond one hospital, we repeated the analyses on a second public dataset, C-NMC 2019 [9], obtained from Kaggle. It contains 10,661 segmented single-cell images (450×450 pixels, cell on a black background) from 73 subjects: 47 with B-lineage acute lymphoblastic leukemia (ALL) and 26 healthy individuals whose normal B cells serve as controls. The images were acquired in a different laboratory, with a different stain and

TABLE I
REPRESENTATIVE PREVIOUS STUDIES. RESULTS ARE AS REPORTED BY EACH STUDY AND ARE NOT DIRECTLY COMPARABLE ACROSS DATASETS. THE LAST COLUMN NOTES AN OPEN QUESTION EACH STUDY WAS NOT DESIGNED TO ADDRESS AND THE CONTRIBUTION OF THIS PAPER (C1–C5) THAT ADDRESSES IT.

Study	Data	Method	Evaluation	Reported result	Open question (addressed by)
Prellberg & Kramer, 2019 [10]	C-NMC, ALL vs. normal cells	ResNeXt with squeeze-excitation	Challenge test set, unseen subjects	Weighted F1 0.889	Cell-level only; missed-case rate and calibration not reported (C3, C5)
Mondal *et al.*, 2021 [11]	C-NMC	Weighted ensemble of five CNNs	Preliminary test set, 28 subjects	F1 0.886; AUC 0.941	Cell-level only; ensemble cost (C3, C5)
Kasim *et al.*, 2025 [12]	ALL-IDB2 + AML-LMU, 390 cells	CNN features + SVM/XGBoost	Image-level split	Accuracy 0.88 (3 classes)	Class and source coincide; patient grouping unavailable (C1, C2)
Jammal & Dahab, 2025 [2]	C-NMC	CNN–ViT with causal reasoning	Split unit not reported	Accuracy 1.00	Transfer to new patients unknown (C1, C2)
Jammal *et al.*, 2025 [3]	ALL image; C-NMC	Neuro-symbolic ViT	5-fold CV, split unit not reported	Accuracy 1.00; 0.97	Transfer to new patients unknown (C1, C2)
Albzour, 2026 [13]	C-NMC	ML, CNN and ViT benchmark	Subject-disjoint, unseen subjects	AUROC 0.913	One cell-level label; no patient-level model (C1, C5)
Matek *et al.*, 2019 [14]	AML-LMU, 200 patients	ResNeXt	Single-cell test set	Expert-level on 15 cell types	Single centre; diagnosis per patient not addressed (C2, C5)
Hehr *et al.*, 2023 [16]	189 patients, 4 AML subtypes	Attention MIL	Patient-level CV	Subtype identification	Single laboratory; operating point not reported (C2, C3)
Sidhom *et al.*, 2021 [19]	APL-AML, 106 patients (*this cohort*)	Single-cell CNN + MIL	Temporal validation	Cell AUC 0.890 → 0.743	Missed APL patients and calibration not reported (C3)
Manescu *et al.*, 2023 [4]	APL-AML (*this cohort*) and others	Annotation-free MIL (VGG-19)	Random 3-fold CV	AUC 0.96	Random rather than temporal split (C1, C2)
de Almeida *et al.*, 2023 [20]	352 + 63 slides	Engineered morphometry + glmnet / MIL	Cross-validation and external cohort	Up to 0.90 AUC (MDS subtype)	Not applied to acute leukemia (C4)
Wei *et al.*, 2025 [17]	572 marrow slides	MIL ensemble	Single-centre test cohort	NPM1 AUC 0.90	External cohort not available (C2)
Koch *et al.*, 2024 [21]	13 datasets, 380k cells	DINOv2 foundation model	Linear probing, MIL	Outperforms generic encoders	Patient-level external evaluation open (C2, C5)
Dasdelen *et al.*, 2026 [25]	6,610 individuals; external incl. this cohort	DinoBloom + transformer MIL	Two external sets	8-class acc. 0.72; malignancy AUC 0.97	Needs large cohort and GPU (C5)
Zamora *et al.*, 2026 [23]	Five datasets	Foundation models + retrieval	Held-out dataset	—	Patient leakage not separated from dataset shift (C1)

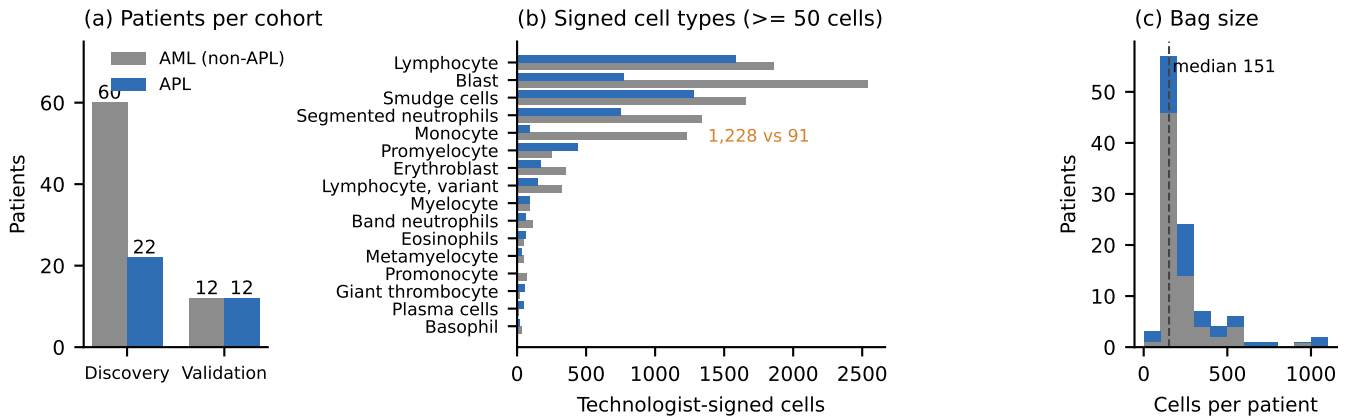


Fig. 1. Cohort composition. (a) Patients per cohort and diagnosis. (b) Labeled cell types with at least 50 cells; monocytes are almost absent in APL (1,228 vs. 91 cells). (c) Number of cells per patient.

microscope, from the APL-AML cohort. The challenge organizers distributed the training data in three subject-disjoint folds. Before any analysis we designated folds 0 and 1 as the development set (42 subjects, 7,108 cells; 30 with ALL) and fold 2 as the external test set (31 subjects, 3,553 cells; 17 with ALL). Every C-NMC cell is labeled with its subject's diagnosis, so the label belongs to the subject, as in T1 below; unlike the APL-AML cells, however, the ALL cells were selected as malignant lymphoblasts, so the label is also partly a property of the cell.

B. Learning Tasks

We defined five tasks (Table III). In T1 every APL-AML cell inherits its patient's diagnosis, which reproduces a common way of building single-cell datasets. In T2 the label is the cell's own type (blast versus other labeled white cells, excluding artefacts, unidentified cells, smudge cells and platelets). T3 is the clinical question: does this patient have APL? T4 and T5

TABLE II
COMPOSITION OF THE APL-AML COHORT.

Cohort	Diagnosis	Patients	Cells	Labeled cells	Cells per patient median (range)
Discovery	AML	60	11,523	8,077	134 (98–984)
	APL	22	8,764	4,082	290 (112–1,066)
Validation	AML	12	3,589	1,893	252 (116–554)
	APL	12	1,848	1,580	140 (83–268)
Total		106	25,724	15,632	152 (83–1,066)

TABLE III
LEARNING TASKS (DEVELOPMENT / EXTERNAL SET). T1–T3: APL-AML (DISCOVERY / VALIDATION); T4, T5: C-NMC 2019 (FOLDS 0–1 / FOLD 2).

Task	Unit	Positive class	Size
T1 Diagnosis on cells	Cell	Cell from an APL patient (43.2%)	20,287 / 5,437 cells
T2 Blast detection	Cell	Labeled blast (28.0%)	9,694 / 2,863 cells
T3 Patient diagnosis	Patient	APL (22 of 82)	82 / 24 patients
T4 C-NMC cells	Cell	Cell from an ALL subject (67.7%)	7,108 / 3,553 cells
T5 C-NMC subjects	Subject	ALL (30 of 42)	42 / 31 subjects

are the C-NMC counterparts of T1 and T3: classifying single cells, and classifying subjects, as ALL or normal.

C. Preprocessing and Cell Segmentation

Fig. 2 outlines the pipeline. A manifest records every image with its patient, cohort, diagnosis and cell label. Unreadable files are kept as missing rows and imputed inside each training fold, and no image is removed because of its label. The Validation cohort is never used for fitting, threshold selection or calibration.

For the engineered features, each image is resized to 192×192 pixels and separated into hematoxylin, eosin and residual channels by color deconvolution [26]. The nucleus is segmented by Otsu thresholding [27] of the smoothed hematoxylin channel, followed by morphological opening and selection of the component nearest the image center. We found that pale cytoplasm could not be separated reliably from neighboring red cells by color, so instead of claiming a full cell outline we describe the cytoplasm with two proxies: a basophilic region of blue-dominant pixels within one nuclear radius, and a perinuclear ring 2–9 pixels wide that excludes the bright background (Fig. 3). C-NMC cells are already segmented and lie on a black background, which the stain-based nucleus segmentation would read as dense chromatin. For C-NMC we therefore set pixels whose brightest channel is below 15 to white before extracting engineered features, so that the same segmentation applies; the CNN received the original images.

D. Feature Representations

The 93 engineered features fall into six groups: nuclear shape (9 features: area, perimeter, circularity, eccentricity, solidity, extent, axis ratio, major axis and number of nuclear fragments); basophilic-region shape, nucleus-to-region ratio and rim size (10); stain and color statistics, i.e., hematoxylin and eosin optical density, CIE $L^*a^*b^*$ and saturation in the nucleus and the ring (26); chromatin texture from gray-level

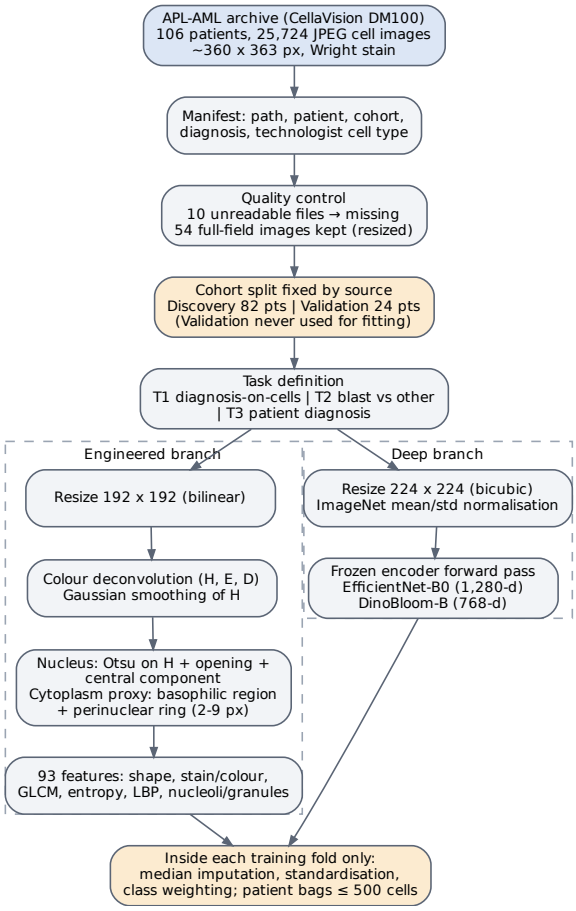


Fig. 2. Preprocessing pipeline. The orange steps prevent leakage: the fixed Discovery/Validation split and fitting every statistic inside the training fold.

co-occurrence matrices [28] at three distances together with nuclear entropy (16); uniform local binary patterns [29] at two radii (28); and counts and areas of pale nuclear spots and dark cytoplasmic granules (4). None of them depends on image size, file format or the source of the image.

As a deep representation we used a frozen EfficientNet-B0 [30] pretrained on ImageNet, which maps each 224×224 input to a 1,280-dimensional vector. The fusion representation concatenates the engineered features and the embedding after column-wise standardization. Because a frozen generic encoder may understate what deep learning can do, we also fine-tuned EfficientNet-B0 on the APL-AML cells. Full fine-tuning was not feasible on our CPU-only workstation, so we froze the stem and the first five stages, stored their output once, and trained the last two stages, the 1,280-channel head and a new classifier, starting from the ImageNet weights. Training used AdamW with a cosine learning-rate schedule, early stopping on validation patients drawn from the training fold, and up to 5,000 training cells per fold (Table IV). The released code also supports the DinoBloom foundation model [21]; it was not evaluated here because its weights could not be downloaded in our computing environment.

On top of these representations we tested feature reduction and selection steps, each fitted inside the training folds only.

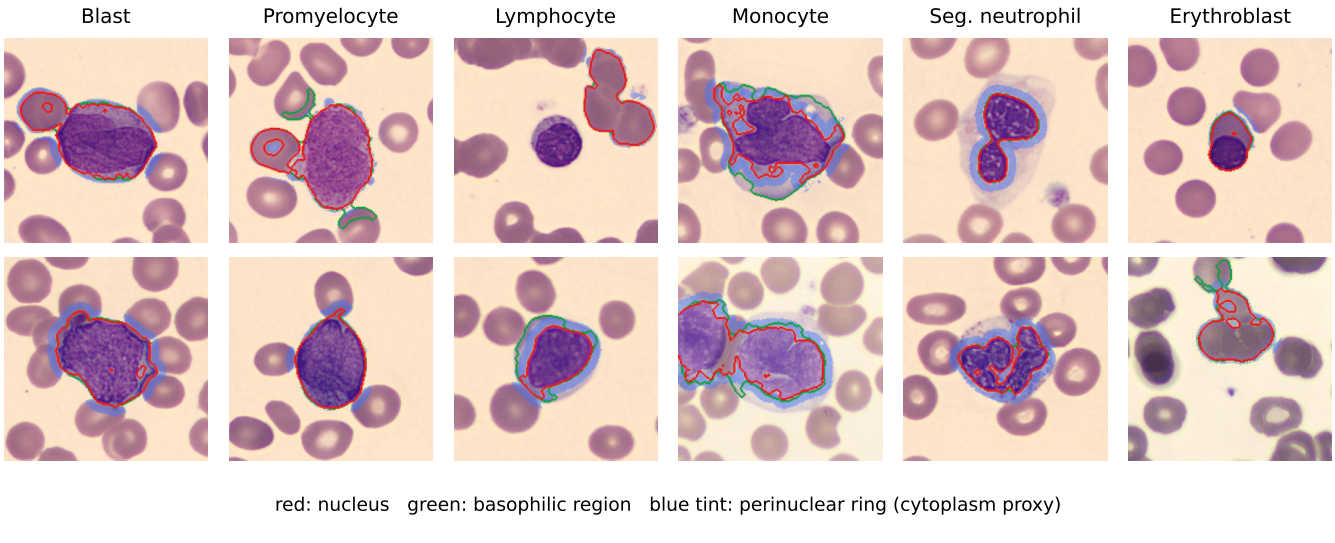


Fig. 3. Segmentation behind the engineered features, shown on cells at the 25th and 75th percentile of each class list. Red: nucleus; green: basophilic region; blue tint: perinuclear ring. The upper lymphocyte and lower erythroblast illustrate failures in which the mask falls on neighboring red cells.

For dimensionality reduction, the cell embeddings $\mathbf{x} \in \mathbb{R}^{1280}$ of the training cells are standardized, $\tilde{\mathbf{x}} = (\mathbf{x} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$, and projected onto the leading k eigenvectors $\mathbf{W}_k \in \mathbb{R}^{1280 \times k}$ of their covariance matrix [31],

$$\mathbf{z} = \mathbf{W}_k^\top \tilde{\mathbf{x}}, \qquad \rho_k = \frac{\sum_{j \le k} \lambda_j}{\sum_j \lambda_j}, \tag{1}$$

where λ_j are the eigenvalues and ρ_k is the fraction of variance retained. We pre-specified $k \in \{16, 32, 64\}$ and added $k \in \{4, 8, 128, 256\}$ as a sensitivity analysis. Because a patient vector holds five statistics per dimension, PCA reduces the patient-level input from 6,400 to $5k$ values. The other steps were a correlation filter that removes the later feature of every pair with $|r| > 0.9$, per-slide standardization of the stain features, L1-penalized selection, and a "digital differential" that describes each patient by the predicted proportions of eight cell classes (Table IV).

E. Classification Models

Fig. 4 shows how the components fit together. At cell level (T1, T2) we used L2-regularized logistic regression (LR) and LightGBM [32], both with class-balanced weights. Each model is a single pipeline that imputes and scales features using training data only.

For patient diagnosis (T3) each patient is represented by up to 500 randomly sampled cells. The pooled baseline summarizes every feature by its mean, standard deviation and 10th, 50th and 90th percentiles and applies LR. The gated attention MIL model [8] maps each cell to $\mathbf{h}_k = \mathrm{ReLU}(\mathbf{W}\mathbf{x}_k)$ with 128 units, weights the cells by

$$a_k = \frac{\exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_k) \odot \sigma(\mathbf{U}\mathbf{h}_k))\}}{\sum_j \exp\{\mathbf{w}^\top(\tanh(\mathbf{V}\mathbf{h}_j) \odot \sigma(\mathbf{U}\mathbf{h}_j))\}}, \tag{2}$$

and classifies the weighted average $\mathbf{z} = \sum_k a_k \mathbf{h}_k$. Because the weights sum to one, the model does not depend on how many cells a patient has. It is trained for 40 epochs with AdamW (learning rate 2×10^{-4}, weight decay 10^{-3}), class-weighted cross-entropy and random removal of 20% of cells as augmentation.

F. Evaluation Protocol and Metrics

Every model was evaluated in three ways (Fig. 5): 5-fold CV on the Discovery cohort with image-level folds, as is common in the literature; 5-fold CV with patient-grouped folds, in which no patient appears in two folds; and a single external test on the Validation cohort. Within each training fold, a further 80/20 split of the same kind was used to fit a calibration temperature [33]. For the external test of T3, the prediction for each Validation patient is the average of all fold models over three seeds, and the decision threshold is chosen on the out-of-fold Discovery predictions of the same models.

Since missing an APL patient can delay life-saving treatment, whereas a false alarm leads only to a confirmatory genetic test, we fixed every decision threshold in advance to reach a sensitivity of at least 0.95 on development data. At that operating point we report sensitivity (recall) together with the number of missed positives, specificity, F1 score, negative predictive value (NPV) and accuracy. AUROC is reported as a threshold-free summary, and calibration as the expected calibration error (ECE, 15 bins). Confidence intervals (95%) come from bootstrap resampling of patients [34], because cells from the same patient are correlated, and differences between two models are tested with a paired patient-level bootstrap. When a table reports several comparisons against the same reference, p-values are also given after Holm correction [35]. C-NMC was analyzed with exactly the same protocol, with folds 0–1 in the role of Discovery and fold 2 in the role of Validation.

Three further analyses address questions that the external tests alone cannot answer. First, because 24 external patients give wide confidence intervals, we repeated the main patient-level comparisons with three repeats of 5-fold patient-level CV over all 106 APL-AML patients, stratified by diagnosis

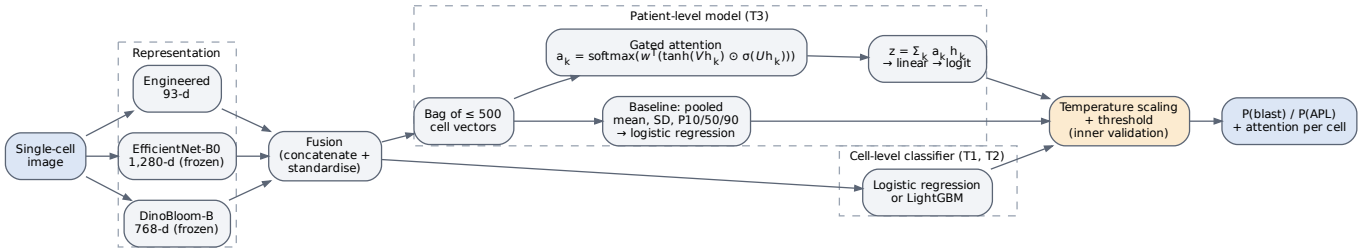


Fig. 4. Model overview. Cells are represented by engineered features, frozen CNN embeddings or both. A cell-level classifier serves T1 and T2; for T3, a patient's cells are aggregated by pooled statistics or by gated attention MIL. Calibration and the decision threshold are fitted on development data only.

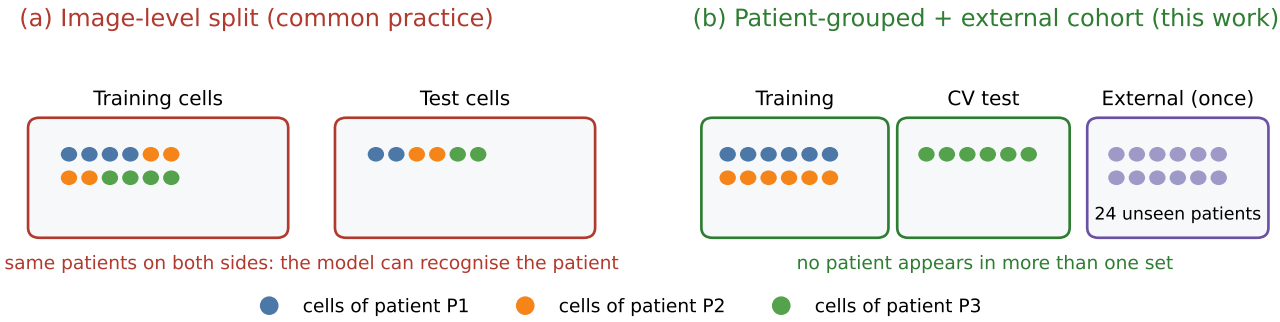


Fig. 5. Evaluation protocols. (a) Image-level splitting places cells of the same patient in training and test sets. (b) Patient-grouped splitting with an external cohort, used in this work; the external cohort is scored once.

and cohort; thresholds were chosen by an inner CV inside each training fold. This analysis has more statistical power but gives up the temporal separation, so it complements rather than replaces the external test. Second, to see whether a new laboratory could restore the intended operating point, we simulated local recalibration: k confirmed local patients ($k/2$ APL, $k/2$ AML, $k = 2, \ldots, 8$) were drawn from the Validation cohort, the threshold was lowered if needed to just below the lowest score of the local APL cases, and the remaining $24 - k$ patients were scored, over 2,000 random draws. Third, decision-curve analysis [36] estimated the net benefit of acting on the model's output at an APL prevalence of 10%, closer to that among patients with suspected acute leukemia than the 50% of the Validation cohort; predicted probabilities were mapped to risks by logistic recalibration on Discovery and shifted to the target prevalence.

G. Implementation Details

All experiments ran on a CPU-only workstation without a GPU (Table V). Feature extraction dominates the cost; once features are stored, each cell-level configuration trains and evaluates in under a minute and each patient-level configuration in about a minute. Table IV lists every setting of every component. All of them were fixed in advance or taken from library defaults; the Validation cohort was scored only after the configuration had been frozen, and no setting was changed afterwards. The attention MIL model is trained for a fixed 40 epochs without early stopping, so no held-out data is consumed for model selection; its training curves are examined in Section IV-F.

IV. Results

A. Exploratory Analysis

Fig. 6 summarizes the exploratory analysis of the engineered features. Every feature group shifted between the two cohorts, and stain and color features shifted most (median standardized mean difference 0.34, Fig. 6a). The shift was large enough to recognize the cohort: among AML patients only, a classifier separated Validation from Discovery cells with an AUROC of 0.75 using stain and color features, compared with 0.59 for shape and texture features and 0.58 for EfficientNet-B0 embeddings (Fig. 6b). Thirty-seven feature pairs were almost redundant ($|r| > 0.9$). The diagnostic signal of individual features was only partly reproducible: patient-level AUROCs in the two cohorts correlated with $r = 0.38$, and 11 of 93 features separated APL from AML in the same direction in both cohorts (Fig. 6d). These observations motivated the feature-engineering experiments below.

B. Effect of the Evaluation Protocol

Table VI and Fig. 7 compare the three protocols. When each cell carried its patient's diagnosis (T1), image-level CV reported AUROCs of 0.938–0.987. Grouping folds by patient lowered them to 0.867–0.901, and the external cohort to 0.658–0.734, so image-level CV overstated external performance by 0.25 to 0.32. When the label described the cell itself (T2), the three protocols nearly agreed and the overstatement was 0.013–0.036. The models themselves changed little: a classifier trained under the leaky protocol scored about the same on the external cohort as one trained under the grouped protocol. What changed was the number reported. The same

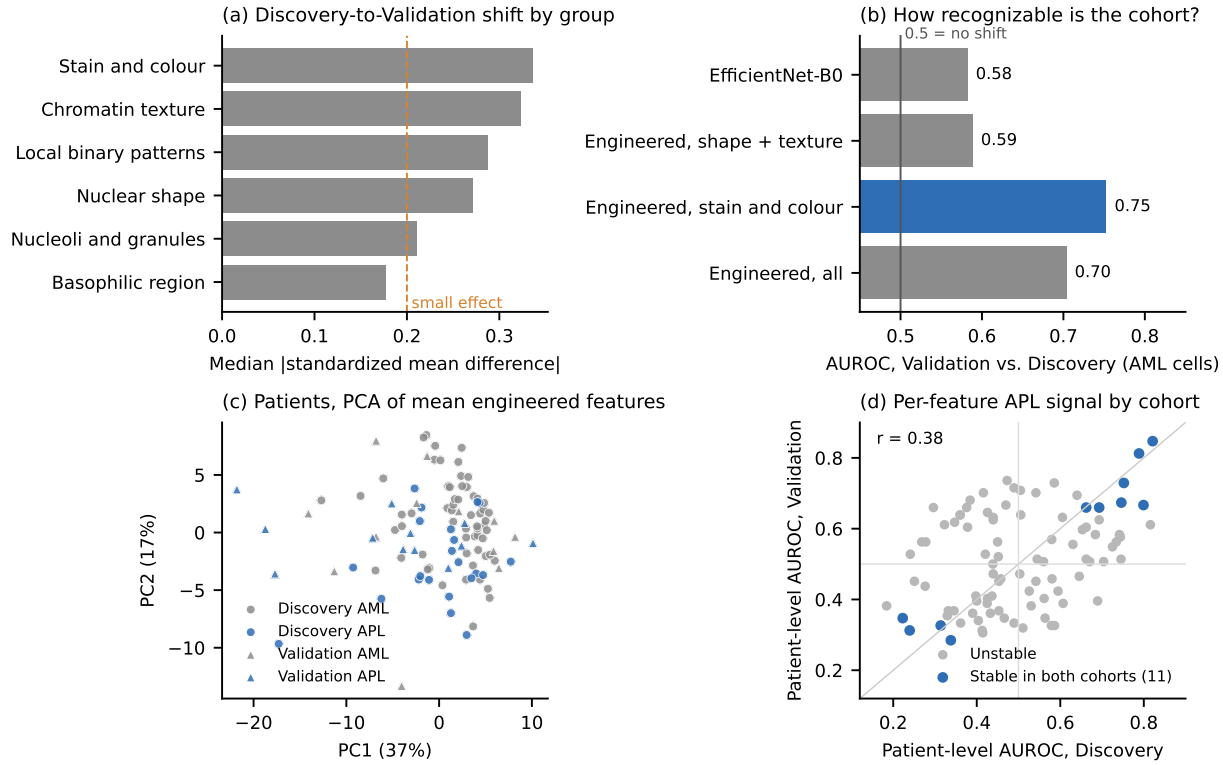


Fig. 6. Exploratory analysis. (a) Shift of engineered feature groups from Discovery to Validation. (b) AUROC of a classifier that tries to tell the two cohorts apart from AML cells; 0.5 means no detectable shift. (c) Patients projected on the first two principal components of their mean engineered features. (d) Patient-level AUROC of each feature for APL vs. AML in the two cohorts; blue points separate the classes consistently.

pattern appeared on C-NMC (Table XIa), where every cell also carries its subject's diagnosis: image-level CV reported 0.912–0.939, subject-grouped CV 0.834–0.864 and the external fold 0.775–0.840, an overstatement of 0.09–0.14. The inflation was smaller than for T1, consistent with C-NMC labels being partly a property of the cell, since the ALL cells were selected as malignant lymphoblasts.

C. Engineered, Frozen and Fine-Tuned Deep Features at Cell Level

For blast detection on new patients, engineered features with LightGBM reached an AUROC of 0.957 (95% CI 0.931–0.977) and detected 99% of blasts at a specificity of 0.62, whereas EfficientNet-B0 embeddings reached 0.920. The paired difference was $+0.034$ (0.008–0.065, $p = 0.002$), and fusing both representations also improved on the embeddings alone ($+0.012$, 0.003–0.024). For diagnosis-on-cells (T1), fusion gave the highest external AUROC among the frozen representations (0.734), although its advantage was not significant. On C-NMC, engineered features again outperformed the frozen embedding on the external fold (0.840 vs. 0.775, Table XIa).

Fine-tuning changed the picture for T1 (Table VII). Adapting the last stages of EfficientNet-B0 raised external AUROC to 0.773 (95% CI 0.612–0.890), the highest of all cell-level models, against 0.694 for the same network used as a frozen feature extractor. The fine-tuned model did not escape the operating-point problem, however: the threshold chosen on Discovery for a sensitivity of 0.95 flagged nearly every Validation cell (recall 0.996, specificity 0.17). ¡¡FT$_T$2 >><< FT_IMAGE >>

D. Feature Engineering

Table VIII reports the feature-engineering experiments for patient-level APL detection. On the Validation cohort, dimensionality reduction had the largest effect. The configuration chosen by the rule fixed in advance—the highest Discovery CV AUROC—was PCA with 64 components, which raised external AUROC from 0.743 to 0.819; 32 components gave the highest external value (0.833), and 16 components the best operating point, detecting 11 of 12 APL patients at a specificity of 0.67. None of these gains was statistically significant, and after Holm correction no feature-engineering step differed significantly from its unreduced reference. Section IV-E shows that the PCA gain did not reproduce in larger or independent data. Removing redundant engineered features changed little. Normalizing stain features within each slide and L1 selection both lowered external AUROC, to 0.58 and 0.59: the color of a patient's cells carries diagnostic information—APL promyelocytes are heavily granulated—that slide-wise normalization removes, and L1 selection with 82 patients chose unstable features. A "digital differential" built from the predicted proportions of eight cell classes reached 0.785 and detected all 12 APL patients, at a specificity of 0.33.

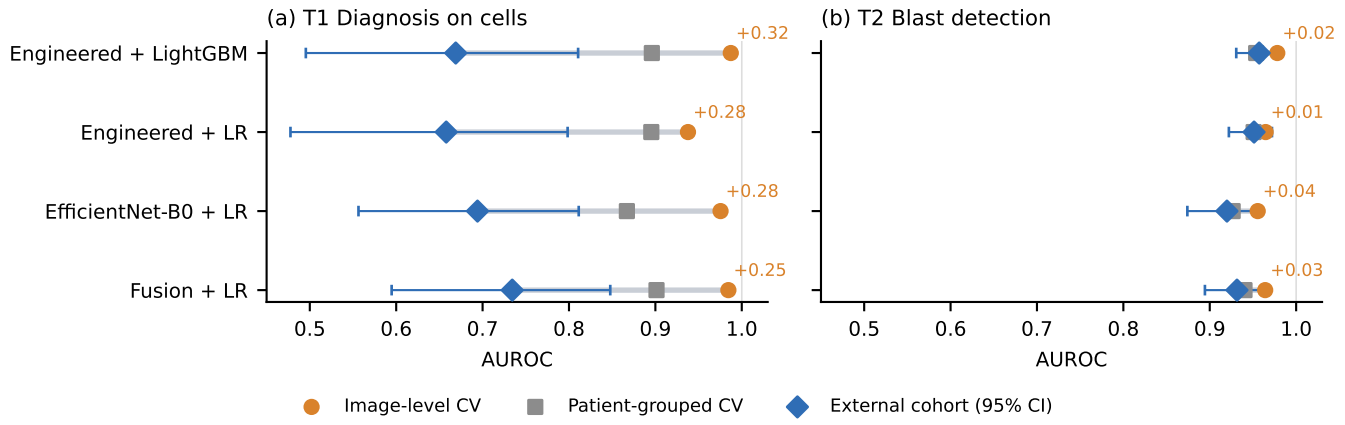


Fig. 7. AUROC under the three protocols (mean of three seeds). Orange numbers give the inflation of image-level CV over the external cohort. (a) Cells labeled with the patient's diagnosis. (b) Cells labeled with their own type.

E. Dimensionality Reduction: Before and After PCA

Table IX isolates the effect of PCA on the pooled patient model built from EfficientNet-B0 embeddings. Without reduction, each APL-AML patient is described by 6,400 values (five statistics of 1,280 dimensions), about 100 for every training patient. This model fitted the Discovery folds well (CV AUROC 0.957) but lost 0.214 of it on the Validation cohort (external AUROC 0.743). Projecting the embeddings onto 32 principal components, which retain 55% of the variance of Discovery cells, shrank the patient vector to 160 values; CV AUROC was unchanged (0.958), external AUROC rose to 0.833, and the gap fell to 0.125. With 16 components the operating point fixed on Discovery detected 11 instead of 6 of the 12 APL patients at the same specificity of 0.67, raising the F1 score from 0.55 to 0.81. Across the component sweep, external AUROC traced an inverted U, peaking between 16 and 64 components (0.79–0.83), while Discovery CV kept rising up to 128 components (Fig. 8a, blue).

These gains did not hold up. First, none was statistically significant: the improvement of PCA-32 over the full embedding was $+0.090$ (95% CI -0.014 to 0.210, $p = 0.10$; Holm-adjusted $p = 0.92$). Second, with more statistical power the advantage disappeared: in pooled CV over all 106 patients, the full embedding ranked patients as well as or better than any reduced version (AUROC 0.953 vs. 0.947 for 32 and 0.942 for 64 components; Section IV-I). Third, on C-NMC PCA was harmful (Fig. 8a, green): the full embedding reached an external AUROC of 0.908, whereas 16, 32 and 64 components reached 0.798, 0.790 and 0.739, and 256 components 0.655 (Holm-adjusted $p < 0.05$ for 128 and 256 components). On both datasets, development CV favored the wrong choice: on C-NMC it increased steadily with the number of components (0.841 to 0.993) while external AUROC fell, so the predeclared selection rule would have picked 128 components and an external AUROC of 0.676.

The diagnostics in Fig. 8c help explain why. We had expected PCA to help by removing acquisition-specific variation, but a classifier that separates the external from the development cohort using cells of the control class did about equally well on every representation in both datasets (APL-AML 0.57–0.64; C-NMC 0.45–0.56), so the leading components did not remove cohort information. The disease signal of single cells, in contrast, grew steadily with the number of components in both datasets, so the discarded components are not noise. Whether a reduced representation transfers better therefore depends on whether the components that matter for the external cohort happen to lie among the leading ones, which no development-only criterion revealed. PCA can limit the size of the patient-level model (Fig. 8b, Section IV-F), but it is not a reliable way to improve generalization, and the number of components should not be tuned on development CV.

F. Training Dynamics and Learning Curves

Fig. 9 shows how the trainable models learned. These curves were computed on the Discovery folds after the configuration had been fixed and were not used to choose epochs, rounds or thresholds. The attention MIL model was trained with a constant learning rate of 2×10^{-4} for 40 epochs. On the full 1,280-dimensional embedding, its training loss fell almost to zero within 10 epochs (0.024). The loss on the held-out fold was lowest at epoch 4 (0.28) and then doubled to 0.60 by epoch 40, while held-out AUROC stayed at 0.96–0.97 (Fig. 9a–c). The model kept ranking patients well but grew increasingly overconfident, in line with its poor external calibration (ECE 0.36, Table X). With PCA-32 inputs, learning was slower and steadier: held-out loss decreased until epoch 31 (0.33) and ended at 0.38, the lowest final value of the three inputs. Engineered features were in between, with a minimum at epoch 10 (0.38) and 0.52 at epoch 40. For all three inputs, held-out AUROC plateaued after about 10 epochs, so the fixed schedule did not cost ranking performance. For the full embedding, however, a shorter schedule would have given better-calibrated probabilities.

LightGBM at cell level (T1, Fig. 9d) behaved similarly. With a learning rate of 0.05, held-out log-loss was lowest after 95 boosting rounds (0.363) and rose to 0.426 at 400 rounds, while training loss kept falling to 0.034.

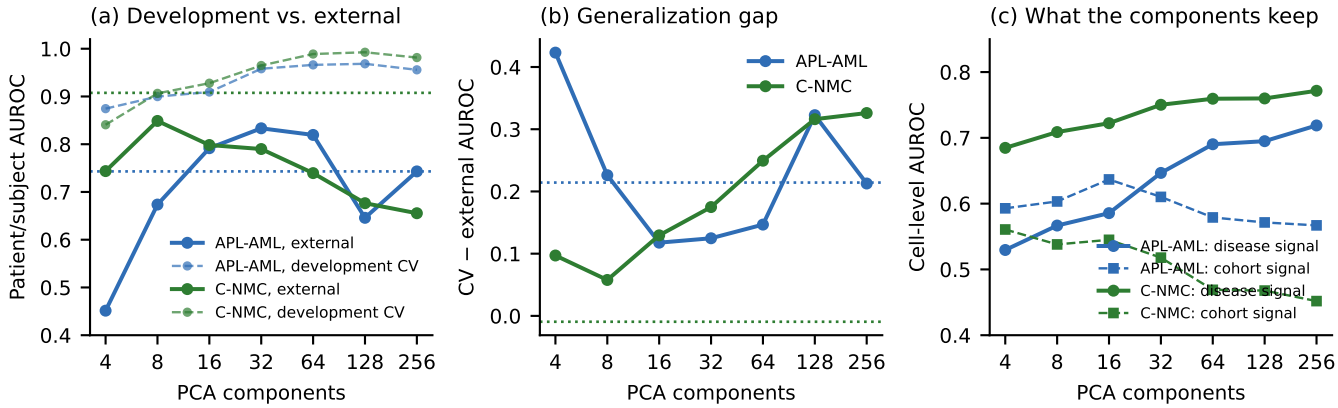


Fig. 8. PCA of EfficientNet-B0 embeddings on both datasets (pooled patient/subject model). (a) AUROC on development CV (dashed) and on the external set (solid) versus the number of components; dotted lines: external AUROC without PCA. (b) Generalization gap (CV minus external AUROC); dotted lines: without PCA. (c) What the components keep: external cell-level disease signal (circles) and cohort signal, i.e., AUROC for telling external from development cells of the control class (squares; 0.5 = none).

The learning curves in Fig. 9e–f show what limits the patient-level models. We trained the pooled model on 16 Discovery patients per fold and up to all of them (65–66). Every representation separated its training patients perfectly at every size (training AUROC 1.00), and held-out-fold AUROC rose to 0.93–0.97 for all three. On the external cohort, however, the curves diverged. PCA-32 was ahead of the full embedding at every training-set size: 0.59 vs. 0.49 with 16 patients, 0.73 vs. 0.64 with 24, 0.80 vs. 0.72 with 48 and 0.82 vs. 0.72 with all patients. With only 24 patients, PCA-32 matched the full embedding trained on all of them, whose curve had flattened after 48 patients, whereas the PCA-32 curve was still rising. On this temporal test, reducing the dimension bought the equivalent of more than twice as many training patients; Section IV-E shows, however, that this advantage did not carry over to pooled CV or to C-NMC.

G. Patient-Level Diagnosis and the Operating Point

Table X lists the patient-level models without feature engineering. Attention MIL on engineered features ranked patients best (external AUROC 0.812, 95% CI 0.614–0.963). The operating point, however, did not transfer. A threshold that detected at least 95% of APL patients in Discovery labeled every Validation patient as APL for this model, and with EfficientNet-B0 features the same rule detected only 6 of 12 APL patients (Fig. 10). Youden thresholds traded the other way, missing half of the APL patients. Even a threshold reset on the Validation cohort itself, which is not possible in practice, could have cleared at most about half of the AML patients without missing an APL patient (oracle specificity 0.42–0.50, Table VIII). The fine-tuned CNN, with cell probabilities averaged per patient as in [19], reached an external AUROC of 0.778 (0.576–0.938) and detected 10 of 12 APL patients at a specificity of 0.67, so the cell-level gain from fine-tuning did not translate into better patient-level diagnosis.

H. Replication on a Second Dataset

Table XIb repeats the patient-level experiment on C-NMC subjects (T5). All six models ranked the 31 external subjects well (AUROC 0.853–0.941), and engineered features again ranked best, both with pooled statistics (0.941, 95% CI 0.840–1.000) and with attention MIL (0.912). In the feature-engineering experiments, engineered features alone reached 0.954–0.958 and their fusion with PCA-32 embeddings 0.966, whereas PCA alone lowered performance (Section IV-E). Unlike on APL-AML, the operating point fixed on development data transferred: every model detected at least 15 of the 17 ALL subjects, three of the six detected all of them, and the pooled models detected 16 or 17 at specificities of 0.71–0.79. The contrast between the two datasets is instructive. On C-NMC the external fold is a random subject-disjoint split of the same collection, whereas the APL-AML Validation cohort was diagnosed later and differs in stain, cell counts and class balance; this suggests that the temporal shift, rather than the models, broke the threshold.

I. Analysis With More Statistical Power

Table XII shows the pooled CV over all 106 APL-AML patients. With 34 APL patients in the test folds, several differences became significant. The unreduced EfficientNet-B0 embedding ranked patients best (AUROC 0.953, 95% CI 0.898–0.989), better than engineered features with pooled statistics (-0.059, Holm-adjusted $p < 0.05$) and better than the promyelocyte fraction used as a manual biomarker (0.677). PCA with 32 or 64 components did not change the ranking (-0.005 and -0.011) but improved specificity at the fixed sensitivity (0.60–0.63 vs. 0.53), and 16 components lost signal (-0.074). Thresholds chosen inside each training fold transferred to the held-out folds (recall 0.89–1.00), which supports the interpretation that the failure on the Validation cohort reflects temporal shift. Because patients from both periods were mixed, this analysis does not measure transfer to a later cohort and should be read alongside, not instead of, the external results.

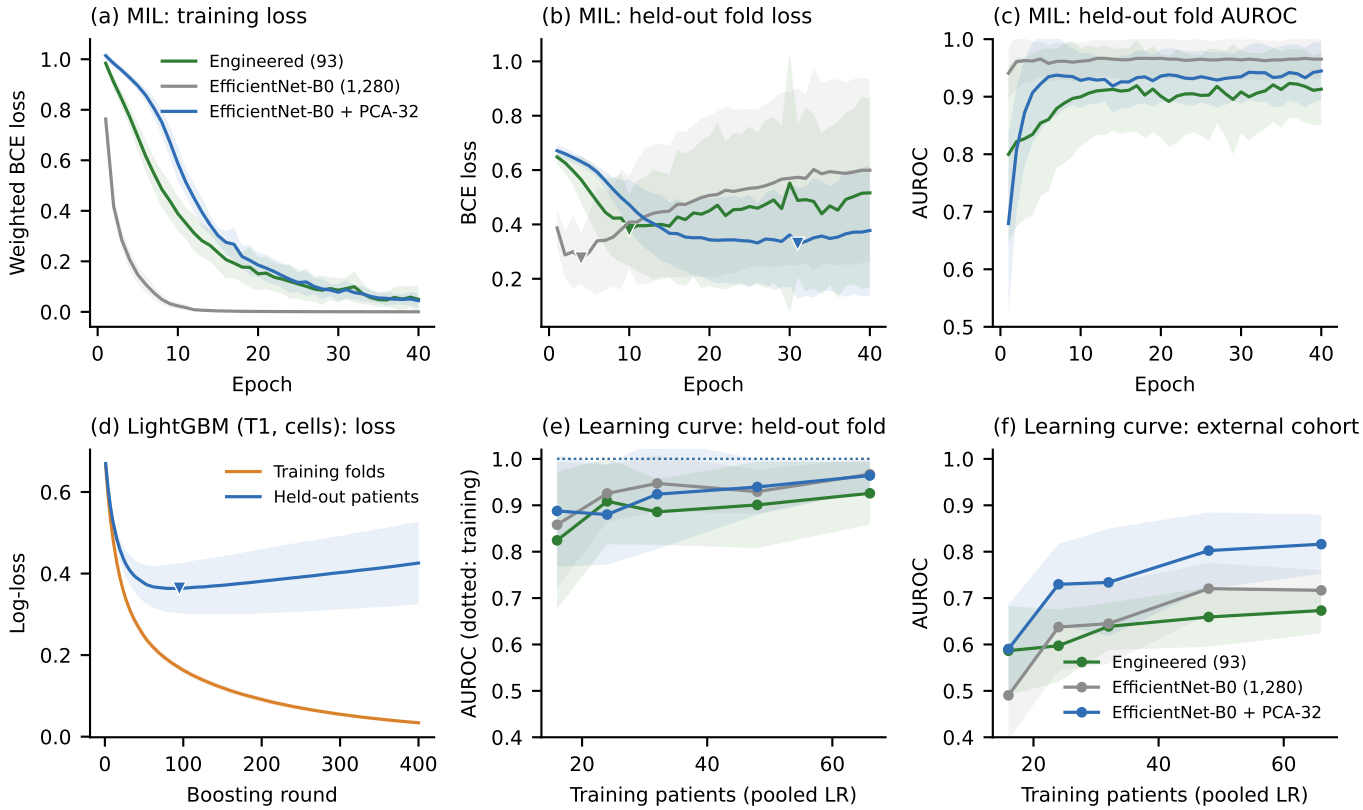


Fig. 9. Training dynamics on the Discovery cohort (mean ± SD over 5 folds × 3 seeds). (a)–(c) Gated attention MIL trained with AdamW at a constant learning rate of 2×10^{-4}: training loss (class-weighted binary cross-entropy), loss and AUROC on the held-out fold per epoch; triangles mark the epoch with the lowest held-out loss. (d) Cell-level LightGBM for T1 (learning rate 0.05): log-loss on training folds and on held-out patients per boosting round. (e), (f) Learning curves of the pooled patient model versus the number of training patients, on the held-out fold (dotted: training AUROC) and on the external cohort. Values in (e) and (f) are for single fold models, not the fold ensemble used in the tables, and are therefore slightly lower.

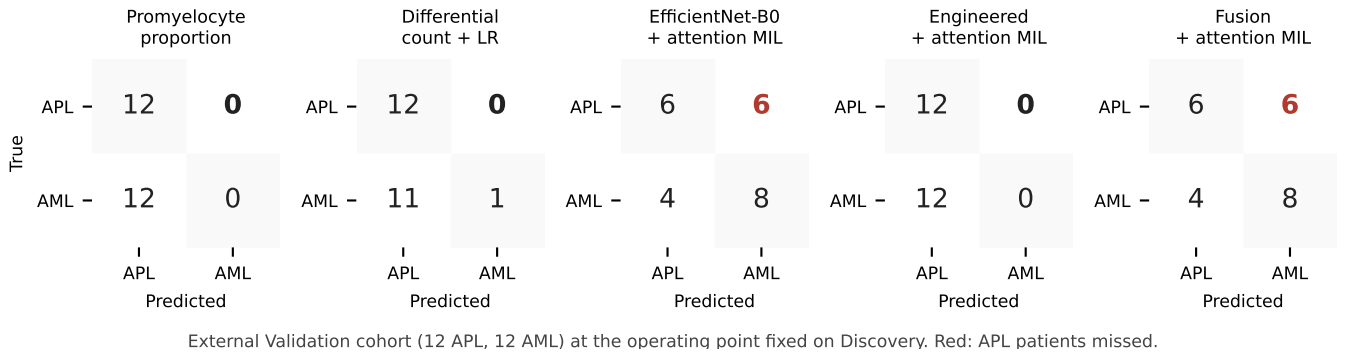


Fig. 10. Confusion matrices on the external cohort at the operating point fixed on Discovery for a sensitivity of at least 0.95. Missed APL patients are shown in red.

J. Can a New Laboratory Restore the Operating Point?

Table XIII and Fig. 11 simulate a laboratory that confirms k local patients before deploying a model. Setting the threshold just below the lowest score of the $m = k/2$ local APL cases gave the same expected recall for every model, about $m/(m+1)$: 0.50 with one local APL case and 0.80 with four (Fig. 11a). This follows from the exchangeability of the local and future APL scores and does not depend on how well the model ranks patients, so about 19 confirmed local APL cases are needed before such a threshold can be expected to reach a recall of 0.95. Combining the development threshold with the local cases—lowering it only when a local APL case would otherwise be missed—raised recall from 0.50 to 0.80 with eight local cases for the frozen CNN models whose development threshold had missed half of the APL patients, at the cost of specificity (from 0.67–0.92 down to 0.44–0.49; Fig. 11b). Even then, the probability of missing at least one APL patient among the remaining 16 was 0.66–0.68. Decision curves at an APL prevalence of 10% (Fig. 11c) showed net benefits close to those of testing every patient at threshold

probabilities up to 5%, and a small positive benefit above that; at a 5% threshold, the attention MIL and fine-tuned models gave net benefits of 0.072 and 0.059 against 0.053 for testing everyone.

K. Comparison With Existing Methods on the Same Cohort

Table XIV compares our models with existing methods on the same patients. The original study reported a cell-level AUC of 0.743 in the later cohort [19], and Manescu *et al.* reported 0.96 with random three-fold CV [4]; neither reported results at a fixed sensitivity. Because the protocols differ, we re-ran representative designs on the identical split with the same operating-point rule. The promyelocyte proportion, the biomarker used as comparator in [19], reached an AUROC of 0.611, and a model on the full differential count reached 0.681. Averaging single-cell predictions per patient, as in [19], reached 0.681–0.799 depending on the features, and the CNN-feature-plus-SVM design of [12] reached 0.806. Our best patient-level models reached 0.812 without feature engineering and 0.833 with PCA. With 24 external patients the confidence intervals of all learned models overlap; the clearest gain is over the manual biomarkers, and the clearest difference between protocols is the gap between random CV (0.96) and temporal validation.

L. Ablation and Interpretation

Table XV and Fig. 12 show which engineered feature groups matter for blast detection. Removing the stain and color group cost -0.034 AUROC (95% CI -0.060 to -0.015), chromatin texture -0.012 and the basophilic region -0.005; nuclear shape, local binary patterns and the nucleolus and granule proxies were redundant given the other groups. Temperature scaling fitted on Discovery left the external calibration error at 0.06–0.13 for cell-level models and 0.26–0.36 for patient-level models. Attention weights of the best patient-level model were highest for cells labeled as monocytes (2.5 times the average in AML patients and 2.8 times in APL patients) and lowest for smudge cells in AML patients (0.31 times the average); since monocytes are nearly absent in APL (Fig. 1b), the model appears to use monocyte-like cells as evidence against APL.

V. Discussion

Four findings stand out. First, the amount of leakage depends on what the label describes, and this held on both datasets. When a label belongs to the patient, every cell carries the same stain, smear and scan date, and a model can learn these instead of the disease. In APL-AML, where scan date and diagnosis are partly confounded [19], the inflation reached 0.32 AUROC. In C-NMC, where the label belongs to the subject but the malignant cells were also selected as such, it was 0.09–0.14. When the label described the cell itself it was at most 0.04. Image-level results for diagnosis-level labels should therefore be read with caution. This may also explain why studies that held out whole subjects, such as the C-NMC challenge entries [10], [11], report F1 scores near 0.89 rather than near 1.0.

Second, ranking transferred better than decisions did, and whether decisions transferred depended on the kind of shift. On a random subject-disjoint split of C-NMC, the threshold fixed for a sensitivity of 0.95 caught 15–17 of 17 ALL subjects; on the later APL-AML cohort, the same rule flagged everyone or missed half of the APL patients, even though the models ranked patients reasonably well. Stain and color, which carry real diagnostic information, drift between the APL-AML cohorts (Fig. 6), and pooled CV without that drift gave thresholds that transferred. Local recalibration is the obvious remedy, but our simulation sets a sobering bound: a threshold placed below the lowest of m local positive cases has an expected recall of only $m/(m+1)$, whatever the model, so about 20 confirmed local cases are needed before a recall of 0.95 can be expected. For a condition in which a missed diagnosis can be fatal, a model like this is best used to prioritize smears for urgent review rather than to rule APL out.

Third, no single representation won everywhere. Engineered features, which are interpretable, were best for blast detection and, alone or fused with CNN features, on both C-NMC tasks; fine-tuning a CNN gave the best single-cell APL classifier on the external cohort; and the unreduced CNN embedding ranked patients best in pooled CV. Removing stain information consistently hurt, because in APL the color of the cytoplasm is part of the disease. Attention MIL ranked patients well but was not significantly better than averaging cell predictions, in line with the modest gains reported for MIL on small cohorts.

Fourth, a result that looked like the clearest improvement in our first analysis did not survive replication. Reducing CNN embeddings to 16–64 principal components raised external AUROC on the 24-patient APL-AML test, but not in pooled CV over 106 patients, and it lowered external AUROC on C-NMC. On both datasets, development CV increased with the number of components while external performance fell beyond some point, so it gave no reliable guide to the dimension. We report this negative result because a single external test of this size can make a chance improvement look like a method, and because dimensionality reduction is often presented as a generic remedy for small medical datasets.

This study has limitations. The two datasets differ in disease (APL vs. other AML; ALL vs. normal B cells), so the replication tests the evaluation principles rather than one clinical model across sites; an independent AML cohort with APL cases remains to be tested. The APL-AML external cohort contains 24 patients and the C-NMC external fold 31 subjects, so patient-level confidence intervals are wide. Cell labels in APL-AML come from an analyzer verified by an operator rather than from expert consensus. Our cytoplasm features rely on proxies, and segmentation fails on some small cells. Fine-tuning was partial (the last two stages), limited to 5,000 cells per fold and run with one seed on a CPU, so a fully fine-tuned or larger network could do better. Hematology foundation models such as DinoBloom [21] were implemented but could not be evaluated in our computing environment.

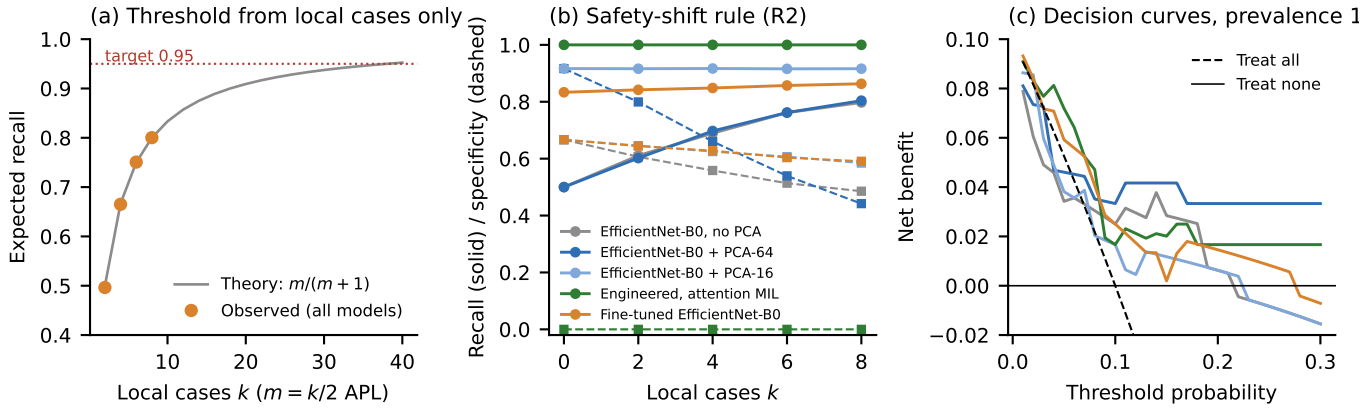


Fig. 11. Restoring the operating point at a new site (APL-AML Validation cohort). (a) Expected recall when the threshold is set just below the lowest score of $m = k/2$ local APL cases: observed values (all models give the same result) and the theoretical $m/(m + 1)$. (b) Recall (solid) and specificity (dashed) when the development threshold is lowered only as far as needed to catch the local APL cases. (c) Decision curves at 10% APL prevalence; dashed line: test every patient; horizontal line: test none.

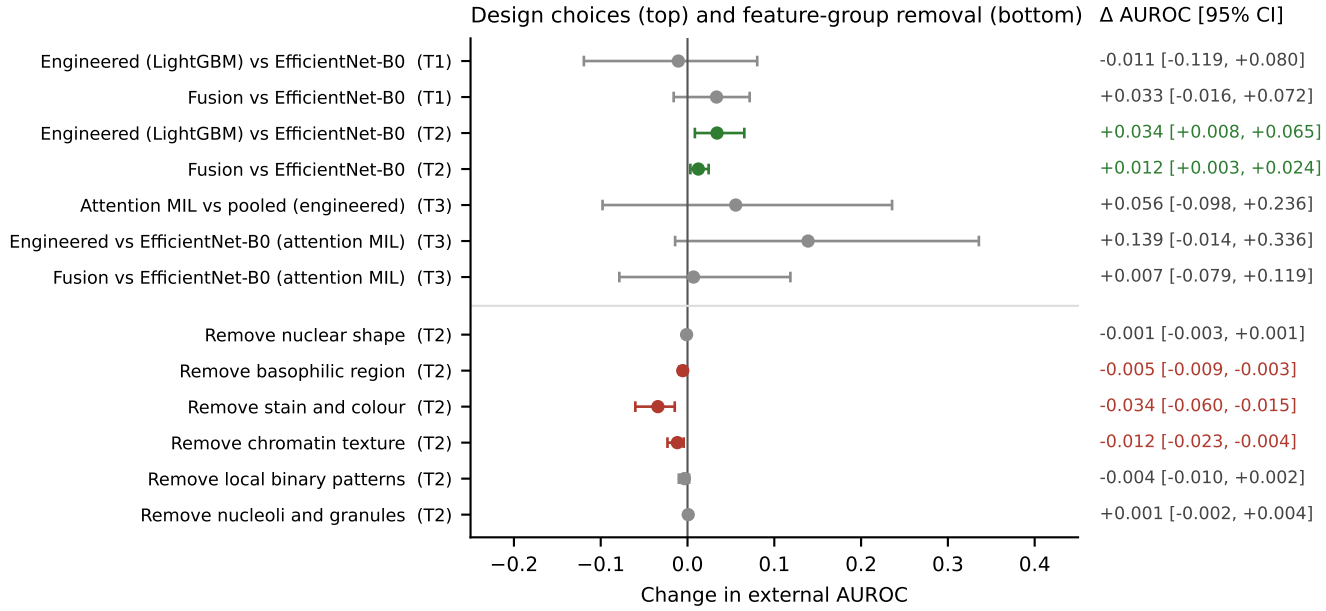


Fig. 12. Effect of design choices (top) and of removing each engineered feature group for blast detection (bottom) on external AUROC. Bars are 95% confidence intervals from a paired patient-level bootstrap.

VI. Conclusion

Reported accuracy for leukemia detection from blood smears depends strongly on how the evaluation is set up. Across two public datasets, splitting cells instead of patients inflated AUROC by up to 0.32 when cells carried the patient's diagnosis. Under patient-level evaluation with external test sets, interpretable engineered features matched or exceeded generic CNN features in most comparisons, fine-tuning helped single-cell classification but not patient diagnosis, and an apparent gain from PCA did not replicate. Decision thresholds transferred to a random held-out split but not to a later cohort, and restoring them locally requires about 20 confirmed positive cases. We recommend that studies in this area split data by patient, state whether labels describe cells or patients, test on more than one external set, report sensitivity, missed cases, specificity, F1 score, NPV and calibration at a pre-specified operating point, and re-check apparent improvements with more data before adopting them.

Data and Code Availability

The APL-AML cohort is available at https://github.com/sidhomj/DeepAPL. Code, split files, label maps and per-model outputs are released at https://github.com/Sherif-Hussein/leukemia-leakage-benchmark. Reporting follows TRIPOD+AI [38].

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TABLE IV
COMPLETE SETTINGS OF EVERY COMPONENT. ALL VALUES WERE FIXED BEFORE THE VALIDATION COHORT WAS SCORED; NONE WAS TUNED ON IT. DEFAULTS OF THE NAMED LIBRARIES APPLY TO ANYTHING NOT LISTED.

Component	Setting	Value
Data and splits	Development / external cohorts	APL-AML: Discovery (82 patients) / Validation (24 patients), split by diagnosis date by the data owners
	Second dataset	C-NMC 2019: official folds 0–1 (42 subjects) / fold 2 (31 subjects), designated before any analysis
	Cross-validation	5 folds; image-level `StratifiedKFold` or patient-level `StratifiedGroupKFold` (cells), `StratifiedKFold` over patients (T3)
	Random seeds	0, 1, 2 (fold assignment, cell sampling, model initialization); results are averaged over seeds
	Cells per patient (T3)	Up to 500, sampled at random once (seed 0); patients with fewer cells keep all
Engineered features (93)	Input	RGB, bilinear resize to 192×192
	Color deconvolution	Ruifrok–Johnston H&E matrix (`skimage rgb2hed`)
	Nucleus	Gaussian smoothing ($\sigma = 1$) of hematoxylin, Otsu threshold, opening (disk 2), component nearest the center (within $0.30\times$ image size); fallback: darkest 15% of gray levels
	Cytoplasm proxies	Basophilic region: blue-dominant pixels within one nuclear radius, opening (disk 2), closing (disk 4); ring: 2–9 px from the nucleus, excluding pixels brighter than the 90th lightness percentile -3
	Texture	GLCM on 33 gray levels, distances 1, 2, 4 px, angles $0, \pi/4, \pi/2, 3\pi/4$ (contrast, dissimilarity, homogeneity, energy, correlation; averaged over angles); uniform LBP with $(P, R) = (8, 1)$ and $(16, 2)$
	Missing values	Unreadable images give NaN rows, imputed with the training-fold median
	C-NMC background	Pixels with brightest channel < 15 set to white before feature extraction (engineered features only)
Deep features	Encoder	EfficientNet-B0, ImageNet weights (`timm`), frozen, global-average-pooled 1,280-d output
	Input	Bicubic resize to 224×224, ImageNet mean/std normalization, no augmentation
	Extraction	Batch size 64, 2 data-loader workers, CPU, `torch.no_grad`
Feature engineering	PCA	Standardization then PCA (`svd_solver`=auto, `random_state`=0), both fitted on training-fold cells only; 16, 32, 64 components pre-specified; 4, 8, 128, 256 as sensitivity analysis
	Correlation filter	Drop the later feature of every pair with $\lvert r \rvert > 0.9$ (training cells, $\leq$20,000 sampled)
	Stain normalization / L1	Per-slide median/SD standardization of the 26 stain features; L1-penalized LR, $C = 0.5$, SAGA solver
Cell-level classifiers (T1, T2)	Logistic regression	L2, $C = 0.1$, L-BFGS, 3,000 iterations, class-balanced weights; median imputation and standardization inside the pipeline
	LightGBM	400 trees, learning rate 0.05, 31 leaves, row subsampling 0.8 (every iteration), column subsampling 0.8, class-balanced weights, 2 threads
	Calibration	Temperature scaling fitted on an inner 80/20 split of the training fold (bounded search 0.05–20)
Pooled patient model (T3)	Patient vector	Mean, SD, 10th, 50th, 90th percentile of every cell feature ($5\times$ dimensions)
	Classifier	LR, L2, $C = 0.1$, 5,000 iterations, class-balanced weights, standardization inside the pipeline
Gated attention MIL (T3)	Architecture	Linear($d \to 128$)–ReLU–Dropout(0.25); gated attention with $\tanh$ and sigmoid branches of 64 units; linear classifier on the attention-weighted mean
	Optimization	AdamW, learning rate 2×10^{-4} (constant, no schedule), weight decay 10^{-3}, one bag per step, 40 epochs, no early stopping
	Loss	Binary cross-entropy with positive-class weight $n_{\mathrm{AML}}/n_{\mathrm{APL}}$ of the training fold
	Augmentation	Random removal of 20% of a bag's cells at every step (bags with more than 10 cells)
	Input scaling	Median imputation and standardization from training-fold cells
Fine-tuned CNN (T1, T2)	Trainable part	EfficientNet-B0 stages 6–7, 1×1 head convolution (1,280 channels), new linear classifier (dropout 0.2); stem and stages 1–5 frozen, their $112 \times 14 \times 14$ outputs cached once (float16)
	Optimization	AdamW, peak learning rate 3×10^{-4}, 1 warm-up epoch then cosine decay, weight decay 10^{-4}, batch 128, 6 epochs, class-weighted binary cross-entropy
	Data	Up to 5,000 training cells per fold (random); 15% of training patients (grouped split) or images (image split) as inner validation for early stopping by validation loss; one seed
	Augmentation	Random horizontal and vertical flips and 90° rotations of the cached feature maps
	B1, B2	Promyelocyte fraction (ranked directly); LR on the 17-class differential

TABLE V
COMPUTING ENVIRONMENT AND RUN TIMES.

Item	Specification
Hardware	Intel Xeon 2.80 GHz, 2 vCPUs, 7.8 GB RAM, no GPU
Software	Ubuntu 24.04.5 LTS; Python 3.13.16; PyTorch 2.14.1 (CPU); scikit-learn 1.9.1 [37]; LightGBM 4.7.0; scikit-image 0.26.0; timm 1.0.30
Random seeds	0, 1, 2
Engineered features	16.1 min for 25,724 cells (2 processes)
EfficientNet-B0 embeddings	51.2 min for 25,724 cells (2 threads)
Cell-level model, 5-fold CV	11–40 s (LR), 15–600 s (LightGBM)
Attention MIL, 5-fold CV	33–61 s per seed

TABLE VI
EFFECT OF THE EVALUATION PROTOCOL AT CELL LEVEL (MEAN OF THREE SEEDS). AUROC UNDER IMAGE-LEVEL CV, PATIENT-GROUPED CV AND ON THE EXTERNAL VALIDATION COHORT; THRESHOLD METRICS ON THE EXTERNAL COHORT AT THE OPERATING POINT FIXED ON DISCOVERY (TARGET SENSITIVITY 0.95). INFLATION = IMAGE-LEVEL CV AUROC MINUS EXTERNAL AUROC.

		AUROC			External cohort, fixed operating point					
Task	Model	Image CV	Grouped CV	External [95% CI]	Recall	Spec.	F1	NPV	Acc.	Inflation
T1 Diagnosis on cells	Engineered + LightGBM	0.987	0.896	0.669 [0.495, 0.811]	0.69	0.55	0.54	0.78	0.60	+0.318
	Engineered + LR	0.938	0.895	0.658 [0.478, 0.798]	0.73	0.46	0.52	0.77	0.55	+0.280
	EfficientNet-B0 + LR	0.975	0.867	0.694 [0.556, 0.811]	0.81	0.43	0.56	0.81	0.56	+0.281
	Fusion + LR	0.985	0.901	**0.734 [0.595, 0.848]**	0.78	0.54	0.59	0.83	0.62	+0.250
T2 Blast detection	Engineered + LightGBM	0.978	0.954	**0.957 [0.931, 0.977]**	0.99	0.62	0.58	1.00	0.70	+0.021
	Engineered + LR	0.965	0.951	0.951 [0.922, 0.972]	0.98	0.70	0.63	0.99	0.76	+0.013
	EfficientNet-B0 + LR	0.956	0.927	0.920 [0.874, 0.956]	0.96	0.64	0.58	0.98	0.70	+0.036
	Fusion + LR	0.964	0.940	0.932 [0.894, 0.961]	0.96	0.68	0.61	0.99	0.74	+0.033

TABLE VII
FINE-TUNED VERSUS FROZEN CNN FEATURES AT CELL LEVEL (APL-AML). THE FINE-TUNED MODEL ADAPTS THE LAST TWO STAGES OF EFFICIENTNET-B0 (TABLE IV); ONE SEED. RECALL, SPECIFICITY AND F1 ON THE EXTERNAL COHORT AT THE OPERATING POINT FIXED ON DISCOVERY.

Task	Model	Image CV	Grouped CV	External [95% CI]	Recall	Spec.	F1
T1	Engineered + LightGBM	0.987	0.896	0.669 [0.495, 0.811]	0.69	0.55	0.54
	Engineered + LR	0.938	0.895	0.658 [0.478, 0.798]	0.73	0.46	0.52
	EfficientNet-B0 + LR	0.975	0.867	0.694 [0.556, 0.811]	0.81	0.43	0.56
	Fusion + LR	0.985	0.901	0.734 [0.595, 0.848]	0.78	0.54	0.59
	Fine-tuned EfficientNet-B0	–	0.875	**0.773 [0.612, 0.890]**	1.00	0.17	0.55
T2	Engineered + LightGBM	0.978	0.954	**0.957 [0.931, 0.977]**	0.99	0.62	0.58
	Engineered + LR	0.965	0.951	0.951 [0.922, 0.972]	0.98	0.70	0.63
	EfficientNet-B0 + LR	0.956	0.926	0.920 [0.874, 0.956]	0.96	0.64	0.58
	Fusion + LR	0.964	0.940	0.931 [0.894, 0.961]	0.96	0.68	0.61

TABLE VIII

FEATURE ENGINEERING FOR PATIENT-LEVEL APL DETECTION. ALL STEPS ARE FITTED INSIDE THE DISCOVERY TRAINING FOLDS; THE VALIDATION COHORT IS SCORED ONCE. Δ: EXTERNAL AUROC CHANGE RELATIVE TO THE UNREDUCED REPRESENTATION (PAIRED PATIENT BOOTSTRAP). THE CONFIGURATION WITH THE HIGHEST DISCOVERY CV AUROC (PRE-DECLARED SELECTION RULE) IS MARKED *. ORACLE COLUMN: SPECIFICITY AT 100% RECALL IF THE THRESHOLD WERE RE-SET ON THE VALIDATION COHORT (REFERENCE ONLY, NOT DEPLOYABLE).

Representation	Engineering step	CV AUROC	External AUROC [95% CI]	Δ [95% CI]	Recall (missed)	Spec.	F1	Oracle spec.
Engineered, all 93	–	0.926	0.729 [0.511, 0.924]	–	0.58 (5/12)	0.67	0.61	0.33
Engineered	Correlation filter ($\|r\| > 0.9$)	0.922	0.736 [0.529, 0.930]	+0.007 [-0.070, +0.084]	0.58 (5/12)	0.67	0.61	0.42
Engineered	Per-slide stain normalization	0.872	0.576 [0.329, 0.818]	-0.153 [-0.326, +0.021]	0.67 (4/12)	0.50	0.62	0.08
Engineered	L1 (LASSO) selection	0.898	0.590 [0.350, 0.820]	-0.139 [-0.274, -0.023]*	0.75 (3/12)	0.42	0.64	0.17
Engineered	Filter + stain normalization	0.869	0.569 [0.322, 0.798]	-0.160 [-0.333, +0.009]	0.67 (4/12)	0.50	0.62	0.08
EfficientNet-B0, 1,280-d	–	0.957	0.743 [0.538, 0.926]	–	0.50 (6/12)	0.67	0.55	0.42
EfficientNet-B0	PCA, 16 components	0.909	0.792 [0.580, 0.963]	+0.049 [-0.096, +0.213]	0.92 (1/12)	0.67	0.81	0.42
EfficientNet-B0	PCA, 32 components	0.958	0.833 [0.644, 0.972]	+0.090 [-0.014, +0.210]	0.50 (6/12)	0.92	0.63	0.42
EfficientNet-B0*	PCA, 64 components	0.966	0.819 [0.604, 1.000]	+0.076 [-0.171, +0.324]	0.50 (6/12)	0.92	0.63	0.00
Fusion	Filter + PCA-32	0.960	0.764 [0.566, 0.938]	+0.021 [-0.105, +0.140]	0.58 (5/12)	0.83	0.67	0.42
Digital differential	8 predicted cell classes	0.852	0.785 [0.570, 0.953]	+0.056 [-0.104, +0.229]	1.00 (0/12)	0.33	0.75	0.50
Differential + engineered	Filtered	0.928	0.764 [0.562, 0.943]	+0.035 [-0.015, +0.111]	0.58 (5/12)	0.67	0.61	0.50
Engineered, attention MIL	Filter + stain normalization	0.922	0.569 [0.326, 0.793]	-0.160 [-0.422, +0.091]	0.42 (7/12)	0.75	0.50	0.17
EfficientNet-B0, attention MIL	PCA-32	0.924	0.799 [0.580, 0.972]	+0.069 [-0.133, +0.281]	0.50 (6/12)	0.75	0.57	0.50

Recall, specificity and F1 at the operating point fixed on Discovery (target sensitivity 0.95). * $p < 0.05$ unadjusted; † $p < 0.05$ after Holm correction across all comparisons in the table.

TABLE IX

PATIENT-LEVEL APL DETECTION BEFORE AND AFTER PCA OF THE EFFICIENTNET-B0 EMBEDDINGS (POOLED-STATISTICS LR). PCA IS FITTED ON THE TRAINING CELLS OF EACH DISCOVERY FOLD. INPUTS: NUMBER OF PATIENT-LEVEL FEATURES (FIVE STATISTICS PER COMPONENT) FOR ABOUT 66 TRAINING PATIENTS PER FOLD. COHORT ID: AUROC FOR TELLING VALIDATION FROM DISCOVERY AML CELLS (0.5 = NO ACQUISITION SIGNAL). CELL APL: EXTERNAL CELL-LEVEL AUROC OF A CELL CLASSIFIER. Δ: CHANGE IN EXTERNAL AUROC RELATIVE TO THE UNREDUCED EMBEDDING (PAIRED PATIENT BOOTSTRAP, 4,000 RESAMPLES). COMPONENTS 16, 32 AND 64 WERE SPECIFIED IN ADVANCE; THE OTHERS FORM A POST HOC SENSITIVITY ANALYSIS.

			Cell level		Patient level, AUROC				External, fixed operating point		
Components	Var.	Inputs	Cohort ID	Cell APL	CV	External [95% CI]	Gap	Δ [95% CI]	Recall (missed)	Spec.	F1
None (1,280)	1.00	6,400	0.583	0.711	0.957	0.743 [0.538, 0.926]	0.214	–	0.50 (6/12)	0.67	0.55
4	0.25	20	0.593	0.530	0.874	0.451 [0.219, 0.700]	0.423	-0.292 [-0.483, -0.119]*†	0.50 (6/12)	0.42	0.48
8	0.35	40	0.603	0.567	0.900	0.674 [0.447, 0.883]	0.226	-0.069 [-0.257, +0.109]	0.50 (6/12)	0.75	0.57
16	0.45	80	0.637	0.586	0.909	0.792 [0.580, 0.963]	0.118	+0.049 [-0.096, +0.213]	0.92 (1/12)	0.67	0.81
32	0.55	160	0.610	0.647	0.958	**0.833 [0.644, 0.972]**	0.125	+0.090 [-0.014, +0.210]	0.50 (6/12)	0.92	0.63
64	0.64	320	0.579	0.690	0.966	0.819 [0.604, 1.000]	0.147	+0.076 [-0.171, +0.324]	0.50 (6/12)	0.92	0.63
128	0.73	640	0.571	0.695	0.968	0.646 [0.400, 0.867]	0.323	-0.097 [-0.313, +0.111]	0.42 (7/12)	0.92	0.56
256	0.83	1,280	0.567	0.719	0.956	0.743 [0.514, 0.937]	0.213	+0.000 [-0.195, +0.203]	0.50 (6/12)	0.92	0.63

Var.: cumulative explained variance on Discovery cells. Gap: CV AUROC minus external AUROC. * $p < 0.05$ unadjusted; † $p < 0.05$ after Holm correction across all comparisons in the table.

TABLE X

PATIENT-LEVEL APL DETECTION (T3). CV: DISCOVERY, MEAN $\pm$ SD OVER THREE SEEDS. EXTERNAL: VALIDATION COHORT (12 APL, 12 AML) AT THE OPERATING POINT FIXED ON DISCOVERY (TARGET SENSITIVITY 0.95); THE LAST THREE COLUMNS USE A YOUDEN THRESHOLD FOR REFERENCE.

			External, sensitivity-first operating point						External, Youden threshold		
Features	Aggregation	CV AUROC	AUROC [95% CI]	Recall (missed)	Spec.	F1	NPV	ECE	Recall (missed)	Spec.	Acc.
Engineered	Pooled + LR	0.867$\pm$0.059	0.757 [0.555, 0.938]	1.00 (0/12)	0.17	0.71	1.00	0.294	0.42 (7/12)	0.92	0.67
EfficientNet-B0	Pooled + LR	0.909$\pm$0.018	0.722 [0.493, 0.914]	0.50 (6/12)	0.67	0.55	0.57	0.338	0.50 (6/12)	0.67	0.58
Fusion	Pooled + LR	0.910$\pm$0.018	0.708 [0.486, 0.909]	0.50 (6/12)	0.67	0.55	0.57	0.339	0.50 (6/12)	0.67	0.58
Engineered	Attention MIL	0.827$\pm$0.003	**0.812 [0.614, 0.963]**	1.00 (0/12)	0.00	0.67	0.00	0.264	0.50 (6/12)	0.75	0.62
EfficientNet-B0	Attention MIL	0.885$\pm$0.021	0.674 [0.444, 0.879]	0.50 (6/12)	0.67	0.55	0.57	0.362	0.50 (6/12)	0.75	0.62
Fusion	Attention MIL	0.886$\pm$0.039	0.681 [0.445, 0.879]	0.50 (6/12)	0.67	0.55	0.57	0.325	0.50 (6/12)	0.75	0.62

TABLE XI
REPLICATION ON THE SECOND DATASET, C-NMC 2019 (ALL VS. NORMAL B CELLS; DEVELOPMENT: OFFICIAL FOLDS 0–1, 42 SUBJECTS; EXTERNAL: FOLD 2, 31 UNSEEN SUBJECTS, 17 WITH ALL). SAME FEATURES, MODELS, PROTOCOL AND OPERATING-POINT RULE AS FOR APL-AML.

(a) Cell level: every cell carries its subject's diagnosis (mean of 3–3 seeds)

Model	Image CV	Grouped CV	External [95% CI]	Recall	Spec.	F1	Inflation
Engineered + LR	0.933	0.864	0.840 [0.730, 0.902]	0.95	0.58	0.89	+0.093
EfficientNet-B0 + LR	0.912	0.834	0.775 [0.640, 0.859]	0.96	0.43	0.87	+0.137
Fusion + LR	0.939	0.864	0.817 [0.692, 0.892]	0.95	0.54	0.88	+0.122

(b) Subject level: is this subject's sample leukemic?

Representation	Aggregation	CV AUROC	External [95% CI]	Recall (missed)	Spec.	F1	NPV
Engineered	Pooled + LR	0.906	0.941 [0.840, 1.000]	0.94 (1/17)	0.79	0.89	0.92
EfficientNet-B0	Pooled + LR	0.855	0.920 [0.807, 0.992]	0.94 (1/17)	0.71	0.86	0.91
Fusion	Pooled + LR	0.878	0.929 [0.824, 0.996]	1.00 (0/17)	0.71	0.89	1.00
Engineered	Attention MIL	0.881	0.912 [0.791, 1.000]	0.88 (2/17)	0.64	0.81	0.82
EfficientNet-B0	Attention MIL	0.874	0.853 [0.704, 0.963]	1.00 (0/17)	0.43	0.81	1.00
Fusion	Attention MIL	0.887	0.853 [0.696, 0.967]	1.00 (0/17)	0.57	0.85	1.00

TABLE XII

SECONDARY ANALYSIS WITH MORE STATISTICAL POWER: REPEATED ($3\times$) 5-FOLD PATIENT-LEVEL CV OVER ALL 106 APL-AML PATIENTS. THIS IS AN INTERNAL ANALYSIS, NOT AN EXTERNAL VALIDATION. Δ: AUROC DIFFERENCE VS. THE UNREDUCED EFFICIENTNET-B0 EMBEDDING (PAIRED PATIENT BOOTSTRAP ON REPEAT-AVERAGED SCORES). RECALL, SPECIFICITY AND F1 AT A THRESHOLD CHOSEN FOR SENSITIVITY ≥ 0.95 INSIDE EACH TRAINING FOLD.

Model	AUROC [95% CI]	Δ vs. no PCA	Recall (missed/34)	Spec.	F1
Promyelocyte fraction (biomarker)	0.677 [0.572, 0.784]	-0.275*†	1.00 (0.0)	0.00	0.49
Engineered, pooled LR	0.893 [0.823, 0.949]	-0.059*†	0.94 (2.0)	0.42	0.59
EfficientNet-B0, no PCA	0.953 [0.898, 0.989]	–	0.95 (1.7)	0.53	0.64
EfficientNet-B0 + PCA-16	0.878 [0.802, 0.941]	-0.074*†	0.98 (0.7)	0.35	0.58
EfficientNet-B0 + PCA-32	0.947 [0.897, 0.985]	-0.005	0.96 (1.3)	0.60	0.69
EfficientNet-B0 + PCA-64	0.942 [0.892, 0.979]	-0.011	0.95 (1.7)	0.63	0.70
Engineered + PCA-32 fusion	0.922 [0.853, 0.973]	-0.030	0.95 (1.7)	0.58	0.67
Engineered, attention MIL	0.867 [0.794, 0.931]	-0.086*	0.89 (3.7)	0.52	0.62

* $p < 0.05$ unadjusted; † after Holm correction.

TABLE XIII

CAN A NEW LABORATORY RESTORE THE OPERATING POINT WITH A FEW LOCAL CASES? k CONFIRMED LOCAL PATIENTS ($k/2$ APL, $k/2$ AML) ARE DRAWN FROM THE VALIDATION COHORT AND THE REMAINING $24 - k$ PATIENTS ARE SCORED (2,000 RANDOM DRAWS). R0: DEVELOPMENT THRESHOLD. R2: DEVELOPMENT THRESHOLD LOWERED, IF NEEDED, TO JUST BELOW THE LOWEST SCORE OF THE LOCAL APL CASES. P(MISS): PROBABILITY THAT AT LEAST ONE APL PATIENT IN THE TEST SET IS MISSED.

	R0, $k = 0$				R2, $k = 4$				R2, $k = 8$			
Model	Recall	Spec.	F1	P(miss)	Recall	Spec.	F1	P(miss)	Recall	Spec.	F1	P(miss)
EfficientNet-B0, no PCA	0.50	0.67	0.55	1.00	0.69	0.56	0.63	0.84	0.80	0.49	0.68	0.68
EfficientNet-B0 + PCA-64 (pre-decl.)	0.50	0.92	0.63	1.00	0.70	0.66	0.68	0.82	0.80	0.44	0.68	0.66
EfficientNet-B0 + PCA-16	0.92	0.67	0.81	1.00	0.92	0.62	0.80	0.83	0.92	0.58	0.79	0.67
Engineered, attention MIL	1.00	0.00	0.67	0.00	1.00	0.00	0.67	0.00	1.00	0.00	0.67	0.00
Fine-tuned EfficientNet-B0	0.83	0.67	0.77	1.00	0.85	0.63	0.76	0.84	0.86	0.59	0.76	0.67

TABLE XIV

COMPARISON WITH EXISTING METHODS ON THE SAME APL-AML COHORT. UPPER BLOCK: RESULTS AS REPORTED BY THE ORIGINAL STUDIES, WHOSE PROTOCOLS DIFFER FROM OURS. LOWER BLOCKS: METHODS RE-RUN ON THE IDENTICAL SPLIT (TRAINED ON DISCOVERY, TESTED ONCE ON THE 24 VALIDATION PATIENTS) WITH THE SAME OPERATING-POINT RULE (SENSITIVITY ≥ 0.95 ON DISCOVERY).

Method	Evaluation	AUROC [95% CI]	Recall (missed)	Spec.	F1	NPV	Acc.
As reported in the literature							
Single-cell CNN, Sidhom *et al.* [19]	Temporal validation, cell level	0.743	–	–	–	–	–
Annotation-free MIL, Manescu *et al.* [4]	Random 3-fold CV, patient level	0.96 ± 0.02	–	–	–	–	–
Re-run on the identical split (Validation cohort, patient level)							
Promyelocyte proportion (biomarker in [19])	Same protocol	0.611 [0.371, 0.837]	1.00 (0/12)	0.00	0.67	0.00	0.50
Differential count + LR (comparator in [25])	Same protocol	0.681 [0.437, 0.886]	1.00 (0/12)	0.08	0.69	1.00	0.54
CNN features + SVM (design of [12])	Same protocol	0.806 [0.615, 0.963]	0.50 (6/12)	0.75	0.57	0.60	0.62
Single-cell classifier, engineered (design of [19])	Same protocol	0.681 [0.445, 0.908]	0.75 (3/12)	0.50	0.67	0.67	0.62
Single-cell classifier, EfficientNet-B0 (design of [19])	Same protocol	0.799 [0.580, 0.972]	0.58 (5/12)	0.92	0.70	0.69	0.75
Single-cell classifier, fusion (design of [19])	Same protocol	0.764 [0.536, 0.957]	0.75 (3/12)	0.75	0.75	0.75	0.75
This work							
Engineered + attention MIL	Same protocol	**0.812 [0.614, 0.963]**	1.00 (0/12)	0.00	0.67	0.00	0.50
EfficientNet-B0 + attention MIL	Same protocol	0.674 [0.444, 0.879]	0.50 (6/12)	0.67	0.55	0.57	0.58
Fusion + attention MIL	Same protocol	0.681 [0.445, 0.879]	0.50 (6/12)	0.67	0.55	0.57	0.58

TABLE XV

FEATURE-GROUP ABLATION ON THE EXTERNAL COHORT (ENGINEERED FEATURES, LR). Δ: CHANGE IN AUROC RELATIVE TO ALL 93 FEATURES, PAIRED PATIENT BOOTSTRAP.

	T2 Blast detection		T1 Diagnosis on cells	
Setting (n)	AUROC	Δ [95% CI]	AUROC	Δ [95% CI]
All features (93)	0.950	–	0.671	–
Without one group				
Nuclear shape (84)	0.949	-0.001 [-0.003, +0.001]	0.681	+0.010 [-0.004, +0.024]
Basophilic region (83)	0.945	-0.005 [-0.009, -0.003]*	0.665	-0.005 [-0.025, +0.010]
Stain and colour (67)	0.916	-0.034 [-0.060, -0.015]*	0.548	-0.123 [-0.272, +0.046]
Chromatin texture (77)	0.939	-0.012 [-0.023, -0.004]*	0.675	+0.004 [-0.007, +0.016]
Local binary patterns (65)	0.947	-0.004 [-0.010, +0.002]	0.670	-0.001 [-0.019, +0.020]
Nucleoli and granules (89)	0.951	+0.001 [-0.002, +0.004]	0.671	+0.000 [-0.006, +0.006]
One group only				
Nuclear shape (9)	0.810	-0.140 [-0.212, -0.082]*	0.549	-0.121 [-0.283, +0.062]
Basophilic region (10)	0.823	-0.127 [-0.198, -0.069]*	0.579	-0.092 [-0.242, +0.073]
Stain and colour (26)	0.912	-0.038 [-0.078, -0.013]*	0.682	+0.011 [-0.032, +0.041]
Chromatin texture (16)	0.802	-0.148 [-0.221, -0.086]*	0.526	-0.145 [-0.331, +0.067]
Local binary patterns (28)	0.793	-0.157 [-0.225, -0.098]*	0.581	-0.090 [-0.241, +0.081]
Nucleoli and granules (4)	0.860	-0.090 [-0.142, -0.054]*	0.500	-0.171 [-0.365, +0.057]

* $p < 0.05$.